

TAIEX Forecasting Using Fuzzy Time Series and Automatically Generated Weights of Multiple Factors

Shyi-Ming Chen, *Fellow, IEEE*, Huai-Ping Chu, and Tian-Wei Sheu

Abstract—In this paper, we present a new method to forecast the Taiwan Stock Exchange Capitalization Weighted Stock Index (TAIEX) using fuzzy time series and automatically generated weights of multiple factors. The proposed method uses the variation magnitudes of adjacent historical data to generate fuzzy variation groups of the main factor (i.e., the TAIEX) and the elementary secondary factors (i.e., the Dow Jones, the NASDAQ and the MIB), respectively. Based on the variation magnitudes of the main factor TAIEX and the elementary secondary factors of a particular trading day, it constructs the occurrence vector of the main factor and the occurrence vectors of the elementary secondary factors on the trading day, respectively. By calculating the correlation coefficients between the numerical data series of the main factor and the numerical data series of each elementary secondary factor, respectively, it calculates the relevance degree between the forecasted variation of the main factor and the forecasted variation of each elementary secondary factor. Based on the correlation coefficients between the numerical data series of the main factor and the numerical data series of each elementary secondary factor on a trading day, it automatically generates the weights of the occurrence vector of the main factor and the occurrence vector of each elementary secondary factor on the trading day, respectively. Then, it calculates the forecasted variation of the main factor and the forecasted variation of each elementary secondary factor on the trading day, respectively, to obtain the final forecasted variation on the trading day. Finally, based on the closing index of the TAIEX on the trading day and the final forecasted variation on the trading day, it generates the forecasted value of the next trading day. The experimental results show that the proposed method outperforms the existing methods.

Index Terms—Correlation coefficients, elementary secondary factors, forecasted variations, fuzzy time series, fuzzy variation groups, main factor, TAIEX, variation magnitude.

I. INTRODUCTION

TIME SERIES have often been used to deal with forecasting problems. Krithikaivasan *et al.* [20] presented a generalized cost-function-based forecasting method for

periodically measured nonstationary traffic. Sfetsos and Siriopoulos [24] presented a method for time series forecasting with a novel clustering algorithm and a pattern recognition scheme. Sfetsos and Siriopoulos [25] pointed out that time series has been a popular tool for analyzing and forecasting a large number of data. They examined three common time series, i.e., the weekly and the biweekly foreign exchange rate, the mean hourly wind speed, and the electric load data, which are formed from the averaging of data collected over a shorter time interval. They presented a method using data of higher frequency to predict the respective number of future data, where the final forecasts are estimated from the averaging of the predicted data. Burr [2] made a comparison of a fuzzy-logic-based method [26] for forecasting an autoregressive time series with a statistically motivated forecasting method.

Song and Chissom [27], [28] proposed the concepts of fuzzy time series. In recent years, some fuzzy forecasting methods [1]–[19], [21], [22], [26]–[35], [37] have been presented based on fuzzy time series. Song and Chissom [28], [29] presented the time-invariant fuzzy time series model and the time-variant fuzzy time series model for forecasting the enrollments of the University of Alabama. Because their models used the max–min composition operations, they required a lot of computation time. Therefore, Chen [3] presented a method by using simple arithmetic operations to overcome the drawback of Song and Chissom's models. Chen and Chang [4] presented a method for multivariable fuzzy forecasting based on fuzzy clustering and fuzzy rule interpolation techniques. Chen and Chen [5] presented a method for forecasting the Taiwan Stock Exchange Capitalization Weighted Stock Index (TAIEX) [40] based on fuzzy time series and fuzzy variation groups. Chen and Chen [6] presented a method to forecast the TAIEX, the enrollments of the University of Alabama and the inventory demand based on high-order fuzzy logical relationships. Chen and Chu [7] presented a method for forecasting the TAIEX based on fuzzy time series and automatically generated weights of defuzzified forecasted fuzzy variations of multiple factors. Chen and Hwang [8] presented a fuzzy time series model, called the two-factor time-variant fuzzy time series model, and developed two algorithms for temperature prediction. Chen and Tanuwijaya [9] presented a multivariate fuzzy forecasting method based on fuzzy time series and automatic clustering techniques. Chen and Tanuwijaya [10] presented a fuzzy forecasting method based on high-order fuzzy logical relationships and automatic clustering techniques. Chen and Wang [11] presented a method for forecasting the TAIEX based on fuzzy-trend logical relationship groups. Chen *et al.* [12] presented a method for forecasting the enrollments of the University of Alabama using automatic clustering techniques and fuzzy logical relationships. Huang [13] presented two kinds of interval definitions (i.e.,

Manuscript received September 19, 2010; revised September 29, 2011; accepted January 7, 2012. Date of publication April 17, 2012; date of current version October 12, 2012. This work was supported in part by the National Science Council, Republic of China, under Grant NSC 100-2221-E-011-118-MY2. This paper was recommended by Associate Editor W. Pedrycz.

S.-M. Chen is with the Department of Computer Science and Information Engineering, National Taiwan University of Science and Technology, Taipei 106, Taiwan, and also with the Graduate Institute of Educational Measurement and Statistics, National Taichung University of Education, Taichung 40306, Taiwan (e-mail: smchen@mail.ntust.edu.tw).

H.-P. Chu is with the Department of Computer Science and Information Engineering, National Taiwan University of Science and Technology, Taipei 106, Taiwan (e-mail: M9315904@mail.ntust.edu.tw).

T.-W. Sheu is with the Graduate Institute of Educational Measurement and Statistics, National Taichung University of Education, Taichung 40306, Taiwan (e-mail: sheu@mail.ntcu.edu.tw).

Digital Object Identifier 10.1109/TSMCA.2012.2190399

the average-based length and the distribution-based length) to deal with forecasting problems based on intervals with different lengths. Huarng [14] presented heuristic models by integrating problem-specific heuristic knowledge with Chen's method [3] to improve the forecasting of the enrollments of the University of Alabama and the Taiwan Futures Exchange (TAIFEX). Huarng and Yu [16] presented a method for dealing with forecasting problems based on the intervals with ratio-based lengths to improve forecasting results. Huarng and Yu [17] presented a hybrid model consisting of a neural network to deal with forecasting problems. Huarng *et al.* [18] integrated a multivariate heuristic function with univariate fuzzy time-series models into multivariate models for improving the forecasting results of the TAIFEX. Hwang *et al.* [19] presented a model to forecast enrollments by using fuzzy time series based on the concept that the enrollment variation of this year is related to the trend of the enrollments of the past years. Lee *et al.* [21] presented a method for temperature prediction and the Taiwan Futures Exchange (TAIFEX) forecasting based on fuzzy logical relationships and genetic algorithms. Song [26] presented sample autocorrelation functions for fuzzy time series model selection. Wang and Chen [30] presented a method for temperature prediction and TAIFEX forecasting based on automatic clustering techniques and two-factors high-order fuzzy time series. Yu [31] presented weighted fuzzy time series models for TAIFEX forecasting. Yu and Huarng [32] applied neural networks to implement a fuzzy time series model to deal with forecasting problems. Yu and Huarng [33] applied neural networks to fuzzy time series forecasting and presented a bivariate fuzzy time series model to forecast the TAIFEX. Kuo *et al.* [35] presented a method for forecasting enrollments based on fuzzy time series and particle swarm optimization techniques. Sullivan and Woodall [36] reviewed the first-order time-invariant fuzzy time series model and the first-order time variant model. They compared these models with a time-invariant Markov model using linguistic labels with probability distributions. Yolcu *et al.* [38] used a single-variable constrained optimization method to determine the ratio for the length of intervals in first-order fuzzy time series models. Moreover, in recent years, some researchers used type-2 fuzzy time series models for stock index forecasting [1], [15]. Bajestani and Zare [1] presented a method to forecast the TAIFEX based on high-order type-2 fuzzy time series. Huarng and Yu [15] presented a method for stock index forecasting based on the type-2 fuzzy time series model. However, the type-2 fuzzy time series models presented in [1] and [15] are too complicated. Moreover, the forecasting accuracy rates of the existing methods [3]–[5], [17], [18], [31], [33] based on type-1 fuzzy time series models are not good enough for forecasting the TAIFEX. Therefore, it is necessary to develop a new method to forecast the TAIFEX.

In this paper, we present a new method to forecast the TAIFEX using fuzzy time series and the automatically generated weights of multiple factors. First, the proposed method uses the variation magnitudes of adjacent historical data to generate fuzzy variation groups of the main factor (i.e., the TAIFEX) and the elementary secondary factors (i.e., the Dow Jones, the NASDAQ and the MIB), respectively. Second, based on the variation magnitudes of the main factor TAIFEX and the elementary secondary factors of a particular trading day, it constructs the occurrence vector of the main factor (i.e., the

TAIFEX) and the occurrence vectors of the elementary secondary factors (i.e., the Dow Jones, the NASDAQ and the MIB) on the trading day, respectively. Third, by calculating the correlation coefficients between the numerical data series of the main factor (constructed from the occurrence vector of the main factor) and the numerical data series of each elementary secondary factor (constructed from the occurrence vector of each elementary secondary factor), respectively, it calculates the relevance degree between the forecasted variation of the main factor and the forecasted variation of each elementary secondary factor. Fourth, based on the correlation coefficients between the numerical data series of the main factor on the trading day and the numerical data series of each elementary secondary factor on the trading day, it automatically generates the weights of the occurrence vector of the main factor and the occurrence vector of each elementary secondary factor on the trading day, respectively. Fifth, it calculates the forecasted variation of the main factor and the forecasted variation of each elementary secondary factor on the trading day, respectively, to get the final forecasted variation. Finally, based on the closing index of the TAIFEX on the trading day and the final forecasted variation on the trading day, it generates the forecasted value of the next trading day. The proposed method gets higher average forecasted accuracy rates than the existing methods. The difference between the proposed method and Chen and Chen's method [5] is that the proposed method is based on automatically generated weights of multiple factors, whereas Chen and Chen's method [5] is based on the statistics of the fuzzy variations appearing in each fuzzy variation group. Moreover, the difference between the proposed method and the methods presented in [1] and [15] is that the proposed method is based on the type-1 fuzzy time series model, whereas the methods presented in [1] and [15] are based on the type-2 fuzzy time series models.

The rest of this paper is organized as follows. In Section II, we briefly review the definitions of fuzzy time series. In Section III, we present a new method for forecasting the TAIFEX using fuzzy time series and automatically generated weights of multiple factors. In Section IV, we make a comparison of the forecasted results of the proposed method with that of the existing methods. The conclusions are discussed in Section V.

II. FUZZY TIME SERIES

The concepts of fuzzy time series were proposed by Song and Chissom [27], [28], based on the fuzzy set theory [39]. Let U be the universe of discourse, where $U = \{u_1, u_2, \dots, u_n\}$. A fuzzy set A in the universe of discourse U is defined as follows:

$$A = f_A(u_1)/u_1 + f_A(u_2)/u_2 + \dots + f_A(u_n)/u_n$$

where f_A is the membership function of the fuzzy set A , $f_A: U \rightarrow [0, 1]$, $f_A(u_i)$ denotes the degree of membership of u_i belonging to the fuzzy set A , $f_A(u_i) \in [0, 1]$ and $1 \leq i \leq n$.

Definition 2.1 [27]: Let $Y(t)$ ($t = \dots, 0, 1, 2, \dots$) be the universe of discourse in which fuzzy sets $f_i(t)$ are defined ($i = 1, 2, \dots$) and let $F(t)$ be a collection of $f_i(t)$ ($i = 1, 2, \dots$). Then, $F(t)$ is called a fuzzy time series on $Y(t)$ ($t = \dots, 0, 1, 2, \dots$).

Definition 2.2 [27]: Let $F(t)$ be caused only by $F(t-1)$. Then, there is a fuzzy relationship between $F(t)$ and $F(t-1)$ expressed by

$$F(t) = F(t-1) \circ R(t, t-1) \quad (1)$$

where $\circ$ denotes the max-min composition operator.

Definition 2.3 [27]: Let $F(t-1) = A_i$ and $F(t) = A_j$, where A_i and A_j are fuzzy sets. The fuzzy logical relationship between $F(t-1)$ and $F(t)$ is denoted by $A_i \rightarrow A_j$, where A_i and A_j are called the left-hand side and the right-hand side of the fuzzy logical relationship, respectively.

Definition 2.4 [3]: Assume that there are the following fuzzy logical relationships with the same left-hand side “ A_i ”, shown as follows:

$$\begin{aligned} A_i &\rightarrow A_{j1}, \\ A_i &\rightarrow A_{j2}, \\ &\vdots \\ A_i &\rightarrow A_{jn}. \end{aligned}$$

Then, these fuzzy logical relationships can be grouped into a fuzzy logical relationship group “Group A_i ”, shown as follows:

$$A_i \rightarrow A_{j1}, A_{j2}, \dots, A_{jn}.$$

Definition 2.5 [22]: Let $F_1(t)$ and $F_2(t)$ be two fuzzy time series. If $F_1(t)$ is caused by $F_1(t-1)$ and $F_2(t-1)$ simultaneously, then this fuzzy logical relationship can be represented by

$$(F_1(t-1), F_2(t-1)) \rightarrow F_1(t)$$

and it is called the two-factors fuzzy time series model, where $F_1(t)$ and $F_2(t)$ are called the main factor and the secondary factor of the two-factors fuzzy time series model, respectively.

III. A NEW METHOD FOR FORECASTING THE TAIEX USING FUZZY TIME SERIES AND AUTOMATICALLY GENERATED WEIGHTS OF MULTIPLE FACTORS

In this section, we present a new method for forecasting the TAIEX using fuzzy time series and automatically generated weights of multiple factors. It is obvious that the variation magnitude of the TAIEX of a trading day is relevant to the variation magnitudes of many factors, including the TAIEX, the Dow Jones, the NASDAQ and the MIB of the previous trading day. In this paper, we let the TAIEX be the main factor and let the Dow Jones, the NASDAQ and the MIB be the elementary secondary factors. For each year, the data from January to October are used as the training data set and the data from November to December are used as the testing data set. The proposed method is now presented as follows.

Step 1) Calculate the variation magnitude of the closing indices between the adjacent historical training data of the main factor and calculate the variation magnitude of the closing indices between the adjacent historical training data of each elementary secondary

factor, respectively, where the variation magnitude VM_t of the closing indices between the adjacent historical training data of each factor on trading day t is calculated as follows:

$$VM_t = \frac{Close_t - Close_{t-1}}{Close_{t-1}} \times 100 \quad (2)$$

where $Close_t$ and $Close_{t-1}$ denote the closing indices on trading day t and trading day $t-1$, respectively. Then, calculate the variation magnitude of the closing indices between the adjacent historical testing data of the main factor and calculate the variation magnitude of the closing indices between the adjacent historical testing data of each elementary secondary factor, respectively.

Step 2) Based on the fluctuation of the variation magnitude of each factor, define the universes of discourse of each factor and partition the universes of discourse of each factor into intervals, respectively. Because the “maximum negative fluctuation” and the “maximum positive fluctuation” of the variation magnitude of the main factor TAIEX are -7% and 7% , respectively, we let the universe of discourse U of the main factor TAIEX be $[-7\%, 7\%]$ and partition the universe of discourse U into fourteen intervals $u_1, u_2, \dots, u_{14}$, where $u_1 = [-7\%, -6\%)$, $u_2 = [-6\%, -5\%)$, $u_3 = [-5\%, -4\%)$, $u_4 = [-4\%, -3\%)$, $u_5 = [-3\%, -2\%)$, $u_6 = [-2\%, -1\%)$, $u_7 = [-1\%, 0\%)$, $u_8 = [0\%, 1\%)$, $u_9 = [1\%, 2\%)$, $u_{10} = [2\%, 3\%)$, $u_{11} = [3\%, 4\%)$, $u_{12} = [4\%, 5\%)$, $u_{13} = [5\%, 6\%)$ and $u_{14} = [6\%, 7\%]$. Because there are no limitations on the variation magnitude of the Dow Jones and because the variation magnitude of the Dow Jones is usually within a specific range, we let the universes of discourse V of the elementary secondary factor Dow Jones be $(-\infty, \infty)$ and partition the universe of discourse V into fourteen intervals $v_1, v_2, \dots, v_{14}$, where $v_1 = (-\infty, -6\%)$, $v_2 = [-6\%, -5\%)$, $v_3 = [-5\%, -4\%)$, $v_4 = [-4\%, -3\%)$, $v_5 = [-3\%, -2\%)$, $v_6 = [-2\%, -1\%)$, $v_7 = [-1\%, 0\%)$, $v_8 = [0\%, 1\%)$, $v_9 = [1\%, 2\%)$, $v_{10} = [2\%, 3\%)$, $v_{11} = [3\%, 4\%)$, $v_{12} = [4\%, 5\%)$, $v_{13} = [5\%, 6\%)$ and $v_{14} = [6\%, \infty)$. Because there are no limitations on the variation magnitude of the NASDAQ and because the variation magnitude of the NASDAQ is usually within a specific range, we let the universes of discourse W of the elementary secondary factor NASDAQ be $(-\infty, \infty)$ and partition the universe of discourse W into fourteen intervals $w_1, w_2, \dots, w_{14}$, where $w_1 = (-\infty, -6\%)$, $w_2 = [-6\%, -5\%)$, $w_3 = [-5\%, -4\%)$, $w_4 = [-4\%, -3\%)$, $w_5 = [-3\%, -2\%)$, $w_6 = [-2\%, -1\%)$, $w_7 = [-1\%, 0\%)$, $w_8 = [0\%, 1\%)$, $w_9 = [1\%, 2\%)$, $w_{10} = [2\%, 3\%)$, $w_{11} = [3\%, 4\%)$, $w_{12} = [4\%, 5\%)$, $w_{13} = [5\%, 6\%)$ and $w_{14} = [6\%, \infty)$. Because there are no limitations on the variation magnitude of the MIB and because the variation magnitude of the elementary secondary factor MIB distributes irregularly, we let the universes of discourse Z of the elementary

secondary factor M1B be $(-\infty, \infty)$ and partition the universe of discourse Z into fourteen intervals $z_1, z_2, \dots$, and z_{14} . By analyzing the historical training data of the universe of discourse Z of the elementary secondary factor M1B, the length of each interval in the universe can be determined, described as follows. First, because the largest variation magnitude between any two consecutive days of the historical training data of the M1B from January 2004 to October 2004 occurred on January 2, 2004, which is equal to -1.085164 , by taking the absolute value R of the largest variation magnitude “ -1.085164 ” as the range of possible changes in either the positive variation magnitude or the negative variation magnitude, we partition the universe of discourse Z into 14 intervals $z_1, z_2, \dots$, and z_{14} , where $z_1 = (-\infty, -1.085164\%)$ and $z_{14} = [1.085164\%, \infty)$. Besides the intervals $z_1 = (-\infty, -1.085164\%)$ and $z_{14} = [1.085164\%, \infty)$, we partition the interval $[-1.085164\%, 1.085164\%)$ in the universe of discourse Z into 12 intervals $z_2, z_3, \dots$, and z_{13} , where the length l of each interval in the interval $[-1.085164\%, 1.085164\%)$ can be calculated, shown as follows:

$$l = \left\lceil \frac{2 \times 1.085164}{12} \right\rceil = 0.180861.$$

Therefore, we can partition the universe of discourse Z of the elementary secondary factor M1B into fourteen intervals $z_1, z_2, \dots$, and z_{14} , where $z_1 = (-\infty, -R\%)$, $z_2 = [-R\%, -5l\%)$, $z_3 = [-5l\%, -4l\%)$, $z_4 = [-4l\%, -3l\%)$, $z_5 = [-3l\%, -2l\%)$, $z_6 = [-2l\%, -l\%)$, $z_7 = [-l\%, 0\%)$, $z_8 = [0\%, l\%)$, $z_9 = [l\%, 2l\%)$, $z_{10} = [2l\%, 3l\%)$, $z_{11} = [3l\%, 4l\%)$, $z_{12} = [4l\%, 5l\%)$, $z_{13} = [5l\%, R\%)$ and $z_{14} = [R\%, \infty)$. That is, $z_1 = (-\infty, -1.085164\%)$, $z_2 = [-1.085164\%, -0.904305\%)$, $z_3 = [-0.904305\%, -0.723444\%)$, $z_4 = [-0.723444\%, -0.542583\%)$, $z_5 = [-0.542583\%, -0.361722\%)$, $z_6 = [-0.361722\%, -0.180861\%)$, $z_7 = [-0.180861\%, 0\%)$, $z_8 = [0\%, 0.180861\%)$, $z_9 = [0.180861\%, 0.361722\%)$, $z_{10} = [0.361722\%, 0.542583\%)$, $z_{11} = [0.542583\%, 0.723444\%)$, $z_{12} = [0.723444\%, 0.904305\%)$, $z_{13} = (0.904305\%, 1.085164\%)$ and $z_{14} = [1.085164\%, \infty)$.

Step 3) Define the linguistic terms $A_1, A_2, A_3, \dots, A_{13}$ and A_{14} of the main factor TAIEX represented by fuzzy sets, respectively, shown as follows:

$$\begin{aligned} A_1 &= 1/u_1 + 0.5/u_2 + 0/u_3 + \dots + 0/u_{12} + 0/u_{13} + 0/u_{14}, \\ A_2 &= 0.5/u_1 + 1/u_2 + 0.5/u_3 + \dots + 0/u_{12} + 0/u_{13} + 0/u_{14}, \\ A_3 &= 0/u_1 + 0.5/u_2 + 1/u_3 + \dots + 0/u_{12} + 0/u_{13} + 0/u_{14}, \\ &\vdots \\ A_{13} &= 0/u_1 + 0/u_2 + 0/u_3 + \dots + 0.5/u_{12} + 1/u_{13} + 0.5/u_{14}, \\ A_{14} &= 0/u_1 + 0/u_2 + 0/u_3 + \dots + 0/u_{12} + 0.5/u_{13} + 1/u_{14}. \end{aligned}$$

Define the linguistic terms $B_1, B_2, B_3, \dots, B_{13}$ and B_{14} of the elementary secondary factor Dow Jones

represented by fuzzy sets, respectively, shown as follows:

$$\begin{aligned} B_1 &= 1/v_1 + 0.5/v_2 + 0/v_3 + \dots + 0/v_{12} + 0/v_{13} + 0/v_{14}, \\ B_2 &= 0.5/v_1 + 1/v_2 + 0.5/v_3 + \dots + 0/v_{12} + 0/v_{13} + 0/v_{14}, \\ B_3 &= 0/v_1 + 0.5/v_2 + 1/v_3 + \dots + 0/v_{12} + 0/v_{13} + 0/v_{14}, \\ &\vdots \\ B_{13} &= 0/v_1 + 0/v_2 + 0/v_3 + \dots + 0.5/v_{12} + 1/v_{13} + 0.5/v_{14}, \\ B_{14} &= 0/v_1 + 0/v_2 + 0/v_3 + \dots + 0/v_{12} + 0.5/v_{13} + 1/v_{14}. \end{aligned}$$

Define the linguistic terms $C_1, C_2, C_3, \dots, C_{13}$ and C_{14} of the elementary secondary factor NASDAQ represented by fuzzy sets, respectively, shown as follows:

$$\begin{aligned} C_1 &= 1/w_1 + 0.5/w_2 + 0/w_3 + \dots + 0/w_{12} + 0/w_{13} + 0/w_{14}, \\ C_2 &= 0.5/w_1 + 1/w_2 + 0.5/w_3 + \dots + 0/w_{12} + 0/w_{13} + 0/w_{14}, \\ C_3 &= 0/w_1 + 0.5/w_2 + 1/w_3 + \dots + 0/w_{12} + 0/w_{13} + 0/w_{14}, \\ &\vdots \\ C_{13} &= 0/w_1 + 0/w_2 + 0/w_3 + \dots + 0.5/w_{12} + 1/w_{13} + 0.5/w_{14}, \\ C_{14} &= 0/w_1 + 0/w_2 + 0/w_3 + \dots + 0/w_{12} + 0.5/w_{13} + 1/w_{14}. \end{aligned}$$

Define the linguistic terms $D_1, D_2, D_3, \dots, D_{13}$, and D_{14} of the elementary secondary factor M1B represented by fuzzy sets, respectively, shown as follows:

$$\begin{aligned} D_1 &= 1/z_1 + 0.5/z_2 + 0/z_3 + \dots + 0/z_{12} + 0/z_{13} + 0/z_{14}, \\ D_2 &= 0.5/z_1 + 1/z_2 + 0.5/z_3 + \dots + 0/z_{12} + 0/z_{13} + 0/z_{14}, \\ D_3 &= 0/z_1 + 0.5/z_2 + 1/z_3 + \dots + 0/z_{12} + 0/z_{13} + 0/z_{14}, \\ &\vdots \\ D_{13} &= 0/z_1 + 0/z_2 + 0/z_3 + \dots + 0.5/z_{12} + 1/z_{13} + 0.5/z_{14}, \\ D_{14} &= 0/z_1 + 0/z_2 + 0/z_3 + \dots + 0/z_{12} + 0.5/z_{13} + 1/z_{14}. \end{aligned}$$

Step 4) Fuzzify the variation magnitude between the adjacent historical training data of the main factor and fuzzify the variation magnitude between the adjacent historical training data of the elementary secondary factors, respectively, to get fuzzy logical relationship groups of the main factor and the elementary secondary factors, respectively. The substeps of Step 4 are shown as follows:

Step 4.1) Fuzzify the variation magnitude of the main factor on each trading day into a fuzzy set defined in Step 3). If the variation magnitude of the main factor on trading day t belongs to u_i , where $1 \leq i \leq 14$, then the variation magnitude of the main factor on trading day t is fuzzified into A_i , where $1 \leq i \leq 14$.

Step 4.2) Fuzzify the variation magnitude of each elementary secondary factor on each trading day into a fuzzy set defined in Step 3. For example, if the variation magnitude of the elementary secondary factor Dow Jones on trading day t belongs to v_j , where $1 \leq j \leq 14$, then the variation magnitude of the elementary secondary factor on trading day t is fuzzified into B_j , where $1 \leq j \leq 14$.

Step 4.3) Group the fuzzy variations of the main factor on the trading days having the same fuzzy variations of the main factor on the previous trading day into a fuzzy variation group. For example, assume that the fuzzy variation of the main factor on trading day t is A_2 and assume that the fuzzy variation of the main factor on trading day $t - 1$ is A_1 ; assume that the fuzzy variation of the main factor on trading day $t + m$ is A_3 and assume that the fuzzy variation of the main factor on trading day $t + m - 1$ is A_1 . Then, the fuzzy variations A_2 and A_3 can be grouped into “Group A_1 ” of the main factor, shown as follows:

Group $A_1 : A_2, A_3$.

Step 4.4) Group the fuzzy variations of the main factor on the trading days having the same fuzzy variation of an elementary secondary factor on the previous trading day into a fuzzy variation group. For example, assume that the fuzzy variation of the main factor on trading day t is A_2 and assume that the fuzzy variation of the elementary secondary factor on trading day $t - 1$ is B_1 ; assume that the fuzzy variation of the main factor on trading day $t + n$ is A_3 and assume that the fuzzy variation of the elementary secondary factor on trading day $t + n - 1$ is B_1 . Then, the fuzzy variations A_2 and A_3 can be grouped into “Group B_1 ” of the elementary secondary factor, shown as follows:

Group $B_1 : A_2, A_3$.

Step 5) According to the fuzzy variations appearing in the fuzzy variation groups of the main factor and the elementary secondary factors obtained in Step 4, construct the fuzzy variation vector of the main factor and the fuzzy variation vectors of the elementary secondary factors, respectively, described as follows. Assume that the fuzzy variation of the main factor on trading day $t - 1$ is A_m and assume that the fuzzy variation of the elementary secondary factor on trading day $t - 1$ is B_s , where m and s are positive integers, $1 \leq m \leq 14$ and $1 \leq s \leq 14$. Let $A_{m1}, A_{m2}, \dots$, and A_{mp} be the fuzzy variations appearing in the fuzzy variation group “Group A_m ” of the main factor and let $N_{m1}, N_{m2}, \dots$, and N_{mp} denote the number of occurrences of fuzzy variations $A_{m1}, A_{m2}, \dots$, and A_{mp} in Group A_m , respectively; let $A_{s(i)1}, A_{s(i)2}, \dots$, and $A_{s(i)q}$ be the fuzzy variations appearing in the fuzzy variation group “Group B_s ” of the i th elementary secondary factor and let $N_{s(i)1}, N_{s(i)2}, \dots$, and $N_{s(i)q}$ denote the number of occurrences of fuzzy variations $A_{s(i)1}, A_{s(i)2}, \dots$, and $A_{s(i)q}$ in Group B_s , respectively, where p and q are positive integers, $1 \leq p \leq 14$ and $1 \leq q \leq 14$. Then, construct the occurrence vector $V_M(t)$ of the main factor and construct the occurrence vector

$V_{S(i)}(t)$ of the i th elementary secondary factor on trading day t , respectively, shown as follows:

$$V_M(t) = [A_{m1} A_{m2} \dots A_{mp} N_{m1} N_{m2} \dots N_{mp}],$$

$$V_{S(i)}(t) = [A_{s(i)1} A_{s(i)2} \dots A_{s(i)q} N_{s(i)1} N_{s(i)2} \dots N_{s(i)q}]$$

where $V_M(t)$ denotes the occurrence vector of the main factor on trading day t derived from the fuzzy variation group “Group A_m ” of the main factor; $V_{S(i)}(t)$ denotes the occurrence vector of the i th elementary secondary factor on trading day t derived from the fuzzy variation group “Group B_s ” of the i th elementary secondary factor. For example, assume that the fuzzy variation of the main factor on trading day $t - 1$ is A_9 and assume that the fuzzy variations appearing in the fuzzy variation group “Group A_9 ” of the main factor TAIEX are $A_8, A_7, A_8, A_9, A_8, A_8, A_7, A_7, A_6, A_6, A_5, A_7, A_9, A_7, A_{13}, A_8, A_5, A_6, A_6, A_7, A_7, A_7, A_7, A_7, A_6, A_{11}, A_7, A_7, A_7, A_8$ and A_{10} , where the number of occurrences of the fuzzy variations $A_5, A_6, A_7, A_8, A_9, A_{10}, A_{11}$ and A_{13} appearing in the fuzzy variation group “Group A_9 ” of the main factor TAIEX are 2, 5, 13, 6, 2, 1, 1 and 1, respectively; assume that the fuzzy variation of the i th elementary secondary factor on trading day $t - 1$ is B_7 and assume that the fuzzy variations appearing in the fuzzy variation group “Group B_7 ” of the i th elementary secondary factor are $A_7, A_8, A_9, A_8, A_7, A_8, A_8, A_9, A_8, A_9, A_7, A_8, A_6, A_8, A_8, A_7, A_6, A_9, A_6, A_8, A_8, A_7, A_{13}, A_8, A_8, A_7, A_5, A_8, A_7, A_3, A_6, A_8, A_{10}, A_5, A_7, A_{10}, A_8, A_8, A_5, A_8, A_6, A_9, A_9, A_6, A_8, A_6, A_8, A_6, A_6, A_7, A_9, A_6, A_7, A_7, A_5, A_7, A_9, A_7, A_7, A_8, A_8, A_7, A_7, A_7, A_8, A_7, A_8, A_8, A_9, A_7, A_9, A_7, A_7, A_7, A_5, A_7, A_8, A_7$ and A_6 , where the number of occurrences of the fuzzy variations $A_3, A_5, A_6, A_7, A_8, A_9, A_{10}$ and A_{13} appearing in the fuzzy variation group “Group B_7 ” of the i th elementary secondary factor are 1, 5, 11, 24, 25, 10, 2 and 1, respectively. Then, we define the occurrence vector $V_M(t)$ of the main factor on trading day t and define the occurrence vector $V_{S(i)}(t)$ of the i th elementary secondary factor on trading day t , respectively, shown as follows:

$$V_M(t) = [A_1 A_2 A_3 A_4 A_5 A_6 A_7 A_8 A_9 A_{10} A_{11} A_{12} A_{13} A_{14} 0 0 0 0 2 5 13 6 2 1 1 0 1 0],$$

$$V_{S(i)}(t) = [A_1 A_2 A_3 A_4 A_5 A_6 A_7 A_8 A_9 A_{10} A_{11} A_{12} A_{13} A_{14} 0 0 1 0 5 11 24 25 10 2 0 0 1 0].$$

Step 6) Calculate the correlation coefficient between the occurrence vector $V_M(t)$ of the main factor on trading day t and the occurrence vector $V_{S(i)}(t)$ of the i th elementary secondary factor on trading day t , respectively, described as follows. Based on the number of occurrences appearing in the occurrence vectors $V_M(t)$ and $V_{S(i)}(t)$ on trading day t , respectively, construct the numerical data series $T_{VM(t)}$ from

the occurrence vector $V_M(t)$ of the main factor on trading day t and construct the numerical data series $T_{VS(i)(t)}$ from the occurrence vector $V_{S(i)}(t)$ of the i th elementary secondary factor on trading day t , respectively. If the number of occurrences of the fuzzy variation A_{mp} appearing in the occurrence vector $V_M(t)$ is N_{mp} , then set the corresponding value in the numerical data series $T_{VM(t)}$ to N_{mp} , where $p = 1, 2, \dots$, and 14. If the number of occurrences of the fuzzy variation $A_{s(i)q}$ appearing in the occurrence vector $V_{S(i)}(t)$ is $N_{s(i)q}$, then set the corresponding value in the numerical data series $T_{VS(i)(t)}$ to $N_{s(i)q}$, where $q = 1, 2, \dots$, and 14. Therefore, the corresponding numerical data series $T_{VM(t)}$ and $T_{VS(i)(t)}$ for the occurrence vectors $V_M(t)$ and $V_{S(i)}(t)$ on trading day t are constructed, respectively, shown as follows:

$$\begin{aligned} T_{VM(t)} &= (N_{m1}, N_{m2}, N_{m3}, N_{m4}, N_{m5}, N_{m6}, N_{m7}, N_{m8}, \\ &\quad N_{m9}, N_{m10}, N_{m11}, N_{m12}, N_{m13}, N_{m14}), \\ T_{VS(i)(t)} &= (N_{s(i)1}, N_{s(i)2}, N_{s(i)3}, N_{s(i)4}, N_{s(i)5}, N_{s(i)6}, \\ &\quad N_{s(i)7}, N_{s(i)8}, N_{s(i)9}, N_{s(i)10}, N_{s(i)11}, \\ &\quad N_{s(i)12}, N_{s(i)13}, N_{s(i)14}). \end{aligned}$$

Then, calculate the correlation coefficient r_i between the numerical data series $T_{VM(t)} = (x_1, x_2, \dots, x_n)$ of the main factor and the numerical data series $T_{VS(i)(t)} = (y_1, y_2, \dots, y_n)$ of the i th elementary secondary factor, respectively, shown as follows [23]:

$$r_i = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} \quad (3)$$

where $\bar{x}$ denotes the mean value of the numerical data series $T_{VM(t)} = (x_1, x_2, \dots, x_n)$ of the main factor, $\bar{y}$ denotes the mean value of the numerical data series $T_{VS(i)(t)} = (y_1, y_2, \dots, y_n)$ of the i th elementary secondary factor, and $r_i \in [-1, 1]$. If r_i is equal to 0, then it means that the forecasted variation of the i th elementary secondary factor is irrelevant to the forecasted variation of the main factor. The larger the positive value of r_i is, the more positive is the relevance between the main factor and the i th elementary secondary factor. The larger the negative value of r_i is, the more negative is the relevance between the main factor and the i th elementary secondary factor. If $r_i > 0$, then we adopt the i th elementary secondary factor to assist the main factor to forecast the variation of the main factor TAIEX of the next trading day. Otherwise, we do not adopt the i th elementary secondary factor to assist the main factor to forecast the variation of the main factor TAIEX of the next trading day.

Step 7) Calculate the weight $W_{VM(t)}$ of the occurrence vector $V_M(t)$ of the main factor on trading day t , and calculate the weight $W_{VS(i)(t)}$ of the occurrence

vector $V_{S(i)}(t)$ of the i th elementary secondary factor on trading day t , respectively, shown as follows:

$$W_{VM(t)} = \frac{1}{1 + \sum_{i=1,2,\dots,n \text{ and } r_i > 0} r_i}, \quad (4)$$

$$W_{VS(i)(t)} = \frac{r_i}{1 + \sum_{i=1,2,\dots,n \text{ and } r_i > 0} r_i} \quad (5)$$

where n denotes the number of elementary secondary factors, r_i denotes the correlation coefficient between the numerical data series $T_{VM(t)}$ (constructed according to the occurrence vector $V_M(t)$ of the main factor on trading day t) and the numerical data series $T_{VS(i)(t)}$ (constructed according to the occurrence vector $V_{S(i)}(t)$ of the i th elementary secondary factor on trading day t), and $1 \leq i \leq n$. If $r_i \leq 0$, it indicates then that there is no correlation or a negative correlation between the numerical data series $T_{VM(t)}$ (constructed according to the occurrence vector $V_M(t)$ of the main factor on trading day t) and the numerical data series $T_{VS(i)(t)}$ (constructed according to the occurrence vector $V_{S(i)}(t)$ of the i th elementary secondary factor on trading day t). In this situation, the occurrence vector $V_{S(i)}(t)$ of the i th elementary secondary factor will be of no help for forecasting the fuzzy variation of the main factor, and we do not adopt this elementary secondary factor to help the main factor for forecasting the variation of the main factor on trading day t . If $r_i > 0$, then according to the correlation coefficient between the numerical data series $T_{VM(t)}$ (constructed according to the occurrence vector $V_M(t)$ of the main factor on trading day t) and the numerical data series $T_{VS(i)(t)}$ (constructed according to the occurrence vector $V_{S(i)}(t)$ of the i th elementary secondary factor on trading day t), we can calculate the weights of the occurrence vectors of the main factor and each elementary secondary factor, respectively. For example, assume that the correlation coefficient between the numerical data series $T_{TAIEX(t)}$ (constructed according to the occurrence vector $V_{TAIEX}(t)$ of the main factor “TAIEX” on trading day t) and the numerical data series $T_{S(D)(t)}$ (constructed according to the occurrence vector $V_{S(D)}(t)$ of the elementary secondary factor “Dow Jones” on trading day t) is 0.9002, assume that the correlation coefficient between the numerical data series $T_{TAIEX(t)}$ (constructed according to the occurrence vector $V_{TAIEX}(t)$ of the main factor “TAIEX” on trading day t) and the numerical data series $T_{S(N)(t)}$ (constructed according to the occurrence vector $V_{S(N)}(t)$ of the elementary secondary factor “NASDAQ” on trading day t) is 0.9718, and assume that the correlation coefficient between the numerical data series $T_{TAIEX(t)}$ (constructed according to the occurrence vector $V_{TAIEX}(t)$ of the main factor “TAIEX” on trading day t) and the numerical data series $T_{S(M)(t)}$ (constructed according to the occurrence vector $V_{S(M)}(t)$ of the elementary

secondary factor “M1B” on trading day t) is 0.8403. Because the correlation coefficients between the numerical data series of the main factor “TAIEX” on trading day t and the numerical data series of each elementary secondary factor on trading day t are all greater than zero, we adopt the elementary secondary factors Dow Jones, NASDAQ and M1B to help the main factor TAIEX for forecasting the variation of the TAIEX. Then, based on (4) and (5), the weight $W_{VM(t)}$ of $V_M(t)$ on trading day t , the weight $W_{VS(D)(t)}$ of $V_{S(D)}(t)$ on trading day t , the weight $W_{VS(N)(t)}$ of $V_{S(N)}(t)$ on trading day t and the weight $W_{VS(M)(t)}$ of $V_{S(M)}(t)$ on trading day t can be calculated, where $W_{VM(t)} = 0.2694$, $W_{VS(D)(t)} = 0.2425$, $W_{VS(N)(t)} = 0.2618$ and $W_{VS(M)(t)} = 0.2264$.

Step 8) Generate the final forecasted variation of the main factor on trading day t . Assume that the occurrence vector $V_M(t)$ of the main factor on trading day t and the occurrence vector $V_{S(i)}(t)$ of the i th elementary secondary factor on trading day t are as follows:

$$V_M(t) = [A_{m1} A_{m2} \dots A_{mp}],$$

$$V_{S(i)}(t) = [A_{s(i)1} A_{s(i)2} \dots A_{s(i)q}].$$

Then, the weight W_{Amh} of the fuzzy variation A_{mh} in $V_M(t)$ and the weight $W_{As(i)k}$ of the fuzzy variation $A_{s(i)k}$ in $V_{S(i)}(t)$ are calculated as follows:

$$W_{Amh} = \frac{N_{mh}}{\sum_{h=1}^p N_{mh}}, \quad (6)$$

$$W_{As(i)k} = \frac{N_{s(i)k}}{\sum_{k=1}^q N_{s(i)k}} \quad (7)$$

where $1 \leq h \leq p$ and $1 \leq k \leq q$. The forecasted fuzzy variation $FVar_M(t)$ of $V_M(t)$ and the forecasted fuzzy variation $FVar_{S(i)}(t)$ of $V_{S(i)}(t)$ are calculated as follows:

$$FVar_M(t) = \sum_{h=1}^p (W_{Amh} \times u_h^M), \quad (8)$$

$$FVar_{S(i)}(t) = \sum_{k=1}^q (W_{As(i)k} \times u_k^M) \quad (9)$$

where u_h^M denotes the midpoint of the interval u_h , the maximum membership value of linguistic term A_{mh} occurs at interval u_h and $1 \leq h \leq p$; u_k^M denotes the midpoint of the interval u_k , the maximum membership value of linguistic term $A_{s(i)k}$ occurs at interval u_k and $1 \leq k \leq q$. For example, assume that the occurrence vector $V_{TAIEX}(t)$ of the main factor TAIEX on trading day t is shown as follows:

$$V_{TAIEX}(t) = [0 \ 0 \ 0 \ 0 \ 2 \ 5 \ 13 \ 6 \ 2 \ 1 \ 1 \ 0 \ 1 \ 0].$$

Then, based on (6), the weights W_{A_5} , W_{A_6} , W_{A_7} , W_{A_8} , W_{A_9} , $W_{A_{10}}$, $W_{A_{11}}$ and $W_{A_{13}}$ of the fuzzy variations A_5 , A_6 , A_7 , A_8 , A_9 , A_{10} , A_{11} and A_{13} can be calculated, respectively, where $W_{A_5} = 0.0645$, $W_{A_6} = 0.1613$, $W_{A_7} = 0.4194$, $W_{A_8} = 0.1935$, $W_{A_9} = 0.0645$, $W_{A_{10}} = 0.0323$, $W_{A_{11}} = 0.0323$ and $W_{A_{13}} = 0.0323$. Based on the results obtained in Step 2, we can see that the midpoints of the intervals u_5 , u_6 , u_7 , u_8 , u_9 , u_{10} , u_{11} and u_{13} are -2.5% , -1.5% , -0.5% , 0.5% , 1.5% , 2.5% , 3.5% and 5.5% , respectively. Then, based on (8), we can see that the forecasted fuzzy variation $Fvar_{TAIEX}(t)$ of the main factor TAIEX on trading day t is calculated as follows:

$$\begin{aligned} Fvar_{TAIEX}(t) &= 0.0645 \times (-2.5\%) + 0.1613 \times (-1.5\%) \\ &\quad + 0.4194 \times (-0.5\%) + 0.1935 \times 0.5\% \\ &\quad + 0.0645 \times 1.5\% + 0.0323 \times 2.5\% + 0.0323 \\ &\quad \times 3.5\% + 0.0323 \times 5.5\% \\ &= -0.0484\%. \end{aligned}$$

Assume that the occurrence vector $V_{S(D)}(t)$ of the elementary secondary factor Dow Jones on trading day t is shown as follows:

$$V_{S(D)}(t) = [0 \ 0 \ 1 \ 0 \ 5 \ 12 \ 24 \ 26 \ 10 \ 2 \ 0 \ 0 \ 1 \ 0].$$

Then, based on (6), the weights W_{A_3} , W_{A_5} , W_{A_6} , W_{A_7} , W_{A_8} , W_{A_9} , $W_{A_{10}}$ and $W_{A_{13}}$ of the fuzzy variations A_3 , A_5 , A_6 , A_7 , A_8 , A_9 , A_{10} and A_{13} can be calculated, respectively, where $W_{A_3} = 0.0127$, $W_{A_5} = 0.0633$, $W_{A_6} = 0.1392$, $W_{A_7} = 0.3038$, $W_{A_8} = 0.3165$, $W_{A_9} = 0.1266$, $W_{A_{10}} = 0.0253$ and $W_{A_{13}} = 0.0127$. Based on the results obtained in Step 2, we can see that the midpoints of the intervals v_3 , v_5 , v_6 , v_7 , v_8 , v_9 , v_{10} and v_{13} are -4.5% , -2.5% , -1.5% , -0.5% , 0.5% , 1.5% , 2.5% and 5.5% , respectively. Then, based on (8), the forecasted fuzzy variation $FVar_{S(D)}(t)$ of the elementary secondary factor Dow Jones on trading day t is calculated as follows:

$$\begin{aligned} FVar_{S(D)}(t) &= 0.0127 \times (-4.5\%) + 0.0633 \times (-2.5\%) \\ &\quad + 0.1392 \times (-1.5\%) + 0.3038 \times (-0.5\%) \\ &\quad + 0.3165 \times 0.5\% + 0.1266 \times 1.5\% + 0.0253 \\ &\quad \times 2.5\% + 0.0127 \times 5.5\% \\ &= -0.0949\%. \end{aligned}$$

Then, based on the weight $W_{VM(t)}$ of the occurrence vector $V_M(t)$ of the main factor on trading day t obtained in Step 7, based on the weight $W_{VS(i)(t)}$ of the occurrence vector $V_{S(i)}(t)$ of the i th elementary secondary factor on trading day t obtained in Step 7, based on the forecasted fuzzy variation $FVar_M(t)$ of the main factor on trading day t and based on the forecasted fuzzy variation $FVar_{S(i)}(t)$ of the i th

elementary secondary factor on trading day t , where $1 \leq i \leq n$ and n denotes the number of elementary secondary factors, calculate the final forecasted variation $FVar_F(t)$ of the main factor on trading day t , shown as follows:

$$FVar_F(t) = W_{VM(t)} \times FVar_M(t) + \sum_{i=1}^n (W_{VS(i)(t)} \times FVar_{S(i)}(t)) \quad (10)$$

where $W_{VM(t)}$ denotes the weight of the occurrence vector $V_M(t)$ of the main factor on trading day t and $FVar_M(t)$ denotes the forecasted fuzzy variation of the main factor on trading day t ; $W_{VS(i)(t)}$ denotes the weight of the occurrence vector $V_{S(i)}(t)$ of the i th elementary secondary factor on trading day t and $FVar_{S(i)}(t)$ denotes the forecasted fuzzy variation of the i th elementary secondary factor on trading day t , where $1 \leq i \leq n$ and n denotes the number of elementary secondary factors.

Step 9) Based on the final forecasted variation $FVar_F(t)$ obtained in Step 8, the forecasted value $F(t)$ of the main factor on trading day t is calculated as follows:

$$F(t) = \begin{cases} Close_{t-1} \times (1 + FVar_F(t)), & \text{if } FVar_F(t) \neq 0 \\ Close_{t-1} \times \left(1 + \frac{FVar_M(t-1)}{3}\right), & \text{otherwise} \end{cases} \quad (11)$$

where $Close_{t-1}$ denotes the closing index of the main factor on trading day $t-1$. In (11), when $FVar_F(t) = 0$, we let $F(t) = Close_{t-1} \times (1 + (FVar_M(t-1)/3))$ due to the fact that the forecasted value $F(t)$ of the main factor on trading day t is relevant to the forecasted fuzzy variation $FVar_M(t-1)$ of the main factor on trading day $t-1$. The experimental results show that when $FVar_F(t) = 0$, we let $F(t) = Close_{t-1} \times (1 + (FVar_M(t-1)/3))$ can get better forecasting results.

Table I shows the actual TAIEX and the forecasted TAIEX (using the main factor TAIEX and the elementary secondary factors Dow Jones, NASDAQ and M1B) from November 2004 to December 2004.

IV. EXPERIMENTAL RESULTS

In this section, we apply the proposed method for forecasting the TAIEX from 1990 to 2004, where we let the TAIEX be the main factor and let the combination of the Dow Jones, the NASDAQ and the M1B be the secondary factors, respectively. The data from January to October for each year are used as the training data, and the data in November and December for each year are used as the testing data. We evaluate the performance of the proposed method using the root-mean-square error (RMSE), which is defined as follows:

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (forecasted\ value_i - actual\ value_i)^2}{n}} \quad (12)$$

TABLE I
ACTUAL TAIEX AND THE FORECASTED TAIEX (USING THE MAIN FACTOR TAIEX AND THE ELEMENTARY SECONDARY FACTORS DOW JONES, NASDAQ AND M1B) IN NOVEMBER AND DECEMBER 2004

Date	Actual Index	Forecasted Index
2004/11/1	5656.17	5718.09
2004/11/2	5759.61	5658.97
2004/11/3	5862.85	5754.24
2004/11/4	5860.73	5863.09
2004/11/5	5931.31	5868.04
2004/11/8	5937.46	5927.67
2004/11/9	5945.20	5942.9
2004/11/10	5948.49	5942.85
2004/11/11	5874.52	5952.69
2004/11/12	5917.16	5882.56
2004/11/15	5906.69	5931.63
2004/11/16	5910.85	5908.03
2004/11/17	6028.68	5912.13
2004/11/18	6049.49	6037.08
2004/11/19	6026.55	6042.75
2004/11/22	5838.42	6010.72
2004/11/23	5851.10	5862.23
2004/11/24	5911.31	5855.49
2004/11/25	5855.24	5911.03
2004/11/26	5778.65	5858.01
2004/11/29	5785.26	5773.75
2004/11/30	5844.76	5783.93
2004/12/1	5798.62	5844.68
2004/12/2	5867.95	5817.26
2004/12/3	5893.27	5861.34
2004/12/6	5919.17	5897.53
2004/12/7	5925.28	5916.83
2004/12/8	5892.51	5910.24
2004/12/9	5913.97	5895.93
2004/12/10	5911.63	5918.24
2004/12/13	5878.89	5915.62
2004/12/14	5909.65	5882.3
2004/12/15	6002.58	5911.86
2004/12/16	6019.23	6002.5
2004/12/17	6009.32	6027.15
2004/12/20	5985.94	6012.86
2004/12/21	5987.85	5991.1
2004/12/22	6001.52	6005.92
2004/12/23	5997.67	6005.86
2004/12/24	6019.42	5997.68
2004/12/27	5985.94	6026.35
2004/12/28	6000.57	5987.87
2004/12/29	6088.49	6018.68
2004/12/30	6100.86	6086.83
2004/12/31	6139.69	6099.45

where n denotes the number of dates needed to be forecasted, “forecasted value _{i} ” denotes the forecasted value of day i , “actual value _{i} ” denotes the actual value of day i and $1 \leq i \leq n$. In Table II, we make a comparison of the RMSE and the average rmse of the proposed method with Huarng *et al.*’s method [18], Chen’s method [3], [33], [34], Huarng and Yu’s method [17], [33], [34] Chen and Chang’s method [4] and Chen and Chen’s method [5]. From Table II, we can see that the proposed method outperforms the existing methods for forecasting the TAIEX from 1999 to 2004. Moreover, we also apply the proposed method to forecast the TAIEX from 1990 to 1999 and compare the forecasted results of the proposed method with Chen and Chen’s method [5] and Yu’s method [31], as shown in Table III. From Table III, we can see that the average RMSEs of the proposed method are smaller than the ones of the methods presented in [5] and [31].

TABLE II
A COMPARISON OF THE RMSEs AND THE AVERAGE rmse FOR DIFFERENT METHODS

Methods \ RMSEs	Year	1999	2000	2001	2002	2003	2004	Average RMSE
Huang et al.'s Method [18] (Use NASDAQ)		N/A	158.7	136.49	95.15	65.51	73.57	105.88
Huang et al.'s Method [18] (Use Dow Jones)		N/A	165.8	138.25	93.73	72.95	73.49	108.84
Huang et al.'s Method [18] (Use MIB)		N/A	169.19	133.26	97.1	75.23	82.01	111.36
Huang et al.'s Method [18] (Use NASDAQ & Dow Jones)		N/A	157.64	131.98	93.48	65.51	73.49	104.42
Huang et al.'s Method [18] (Use NASDAQ & MIB)		N/A	155.51	128.44	97.15	70.76	73.48	105.07
Huang et al.'s Method [18] (Use NASDAQ & Dow Jones & MIB)		N/A	154.42	124.02	95.73	70.76	72.35	103.46
Chen's Fuzzy Time Series Model (U_FTS Model) [3], [33], [34]		120	176	148	101	74	84	117.4
Univariate Conventional Regression Model (U_R Model) [30], [31]		164	420	1070	116	329	146	374.2
Univariate Neural Network Model (U_NN Model) [33], [34]		107	309	259	78	57	60	145.0
Univariate Neural Network-Based Fuzzy Time Series Model (U_NN_FTS Model) [17], [33], [34]		109	255	130	84	56	116	125.0
Univariate Neural Network-Based Fuzzy Time Series Model with Substitutes (U_NN_FTS_S Model) [17], [33], [34]		109	152	130	84	56	116	107.8
Bivariate Conventional Regression Model (B_R Model) [33], [34]		103	154	120	77	54	85	98.8
Bivariate Neural Network Model (B_NN Model) [33], [34]		112	274	131	69	52	61	116.4
Bivariate Neural Network-Based Fuzzy Time Series Model (B_NN_FTS model) [33], [34]		108	259	133	85	58	67	118.3
* Bivariate Neural Network-Based Fuzzy Time Series Model with Substitutes (B_NN_FTS_S Model) [33], [34]		112	131	130	80	58	67	96.4
Chen and Chang's Method [4]	Use NASDAQ	123.64	131.10	115.08	73.06	66.36	60.48	94.95
	Use Dow Jones	101.97	148.85	113.70	79.81	64.08	82.32	98.46
	Use MIB	156.92	142.70	132.76	96.06	90.27	100.10	119.80
	Use Dow Jones & NASDAQ	106.34	130.13	113.33	72.33	60.29	68.07	91.75
	Use NASDAQ & MIB	116.22	134.63	116.59	76.48	53.51	69.29	94.45
	Use NASDAQ & Dow Jones & MIB	111.70	129.42	113.67	66.82	56.10	64.76	90.41
Chen and Chen's Method [5]	Use Dow Jones	115.47	127.51	121.98	74.65	66.02	58.89	94.09
	Use NASDAQ	119.32	129.87	123.12	71.01	65.14	61.94	95.07
	Use MIB	120.01	129.87	117.61	85.85	63.1	67.29	97.29
	Use Dow Jones & NASDAQ	116.64	123.62	123.85	71.98	58.06	57.73	91.98
	Use Dow Jones & MIB	116.59	127.71	115.33	77.96	60.32	65.86	93.96
	Use NASDAQ & MIB	114.87	128.37	123.15	74.05	67.83	65.09	95.56
The Proposed Method	Use NASDAQ & Dow Jones & MIB	112.47	131.04	117.86	77.38	60.65	65.09	94.08
	Use TAIEX	100.83	122.20	120.30	72.23	56.89	55.40	87.98
	Use Dow Jones	99.87	122.75	117.18	68.45	53.96	52.55	85.79
	Use NASDAQ	102.6	119.98	114.81	69.07	53.16	53.57	85.53
	Use MIB	101.22	123.99	117.75	70.63	54.92	55.29	87.3
	Use Dow Jones & NASDAQ	101.33	121.27	114.48	67.18	52.72	52.27	84.88
	Use Dow Jones & MIB	100.59	124.1	116.28	68.11	53.5	53.33	85.99
	Use NASDAQ & MIB	102.25	122.47	115.02	68.51	52.82	53.99	85.84
	Use NASDAQ & Dow Jones & MIB	101.47	122.88	114.47	67.17	52.49	52.84	85.22

(*Note: In [34], Yu and Huang have corrected the typing errors appearing in Table VI of [33].)

V. CONCLUSION

In this paper, we have presented a new method for forecasting the TAIEX based on fuzzy time series and the automatically generated weights of multiple factors. First, the proposed method uses the variation magnitudes of adjacent historical data to generate fuzzy variation groups of the main factor (i.e., the TAIEX) and the elementary secondary factors (i.e., the Dow Jones, the NASDAQ and the MIB), respectively. Then, based on the variation magnitudes of the main factor TAIEX and the elementary secondary factors of a particular trading day, it constructs the occurrence vector of the main factor

(i.e., the TAIEX) and the occurrence vectors of the elementary secondary factors (i.e., the Dow Jones, the NASDAQ and the MIB) on the trading day, respectively. By calculating the correlation coefficients between the numerical data series of the main factor (constructed from the occurrence vector of the main factor) and the numerical data series of each elementary secondary factor (constructed from the occurrence vector of each elementary secondary factor), respectively, it calculates the relevance degree between the forecasted variation of the main factor and the forecasted variation of each elementary secondary factor. Then, based on the correlation coefficients

TABLE III
A COMPARISON OF THE RMSEs AND THE AVERAGE RMSE FOR DIFFERENT METHODS (FROM 1990 TO 1999)

RMSEs		Year										
Methods		1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	Average RMSE
Conventional Models [31]	Average-Based Lengths	220	80	60	110	112	79	54	148	167	149	117.9
	Distribution-based lengths	270	79	60	105	132	79	52	149	159	159	124.4
Weighted Models [31]	Average-Based Lengths	227	61	67	105	135	70	54	133	151	142	114.5
	Distribution-Based Lengths	266	67	56	105	114	70	52	152	154	145	118.1
Chen and Chen's Method [5]	Use Dow Jones	172.89	72.87	43.44	103.21	78.63	66.66	59.75	139.68	124.44	115.47	97.70
	Use NASDAQ	169.93	66.12	49.61	104.75	75.66	67.01	60.90	140.86	144.13	119.32	99.83
	Use Dow Jones & NASDAQ	172.99	74.85	43.78	101.38	78.13	68.14	61.26	139.29	132.94	116.64	98.94
The Proposed Method	Use Dow Jones	175.67	43.31	42.82	104.34	94.27	54.0	50.47	138.31	120.0	99.87	92.31
	Use NASDAQ	176.6	43.73	42.42	104.18	95.04	54.49	49.84	139.85	119.22	102.6	92.8
	Use Dow Jones & NASDAQ	174.62	43.22	42.66	104.17	94.6	54.24	50.5	138.51	117.87	101.33	92.17

between the numerical data series of the main factor on the trading day and the numerical data series of each elementary secondary factor on the trading day, it automatically generates the weights of the occurrence vector of the main factor and the occurrence vector of each elementary secondary factor on the trading day, respectively. Then, it calculates the forecasted variation of the main factor and the forecasted variation of each elementary secondary factor on the trading day, respectively, to get the final forecasted variation. Finally, based on the closing index of the TAIEX on the trading day and the final forecasted variation on the trading day, it generates the forecasted value of the next trading day. From Table II and Table III, we can see that the proposed method gets higher average forecasting accuracy rates than the existing methods. Song [26] has presented a method for fuzzy time series model selection with sample autocorrelation functions to choose a model of proper order according to the sample autocorrelation function values. In the future, we will propose a method to automatically determine the optimal order of the high-order fuzzy time series to handle forecasting problems.

ACKNOWLEDGMENT

The authors would like to thank the referees for providing very helpful comments and suggestions. Their insight and comments led to a better presentation of the ideas expressed in this paper. The authors would also like to thank Mr. C.-Y. Wang for his help during this work.

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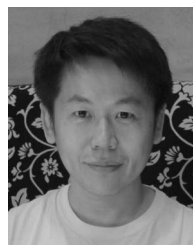
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Shyi-Ming Chen (M'88–SM'96–F'11) received the Ph.D. degree in electrical engineering from National Taiwan University, Taipei, Taiwan, in June 1991. From August 1987 to July 1989 and from August 1990 to July 1991, he was an Instructor in the Department of Electronic Engineering, Fu-Jen Catholic University, New Taipei City, Taiwan. From August 1991 to July 1996, he was an Associate Professor in the Department of Computer and Information Science, National Chiao Tung University, Hsinchu, Taiwan. From August 1996 to July 1998, he was a

Professor in the Department of Computer and Information Science, National Chiao Tung University, Hsinchu, Taiwan. From August 1998 to July 2001, he was a Professor in the Department of Electronic Engineering, National Taiwan University of Science and Technology (NTUST), Taipei, Taiwan. Since August 2001, he has been a Professor in the Department of Computer Science and Information Engineering, National Taiwan University of Science and Technology, Taipei, Taiwan. From August 2007 to July 2010, he was the Dean of the College of Electrical Engineering and Computer Science, Jinwen University of Science and Technology, New Taipei City, Taiwan. Since August 2011, he has been the Vice President of the National Taichung University of Education, Taichung, Taiwan. He was a Visiting Scholar in the Department of Electrical Engineering and Computer Science, University of California, Berkeley, in 1999. He was a Visiting Scholar in the Institute of Information Science, Academia Sinica, Taipei, Taiwan, in 2003 and 2011. He has published more than 390 papers in referred journals, conference proceedings and book chapters. His research interests include fuzzy systems, information retrieval, knowledge-based systems, artificial intelligence, neural networks, data mining and genetic algorithms.

Prof. Chen is a Fellow of the Institution of Engineering and Technology (IET), a Fellow of the International Fuzzy Systems Association (IFSA), and a member of the Phi Tau Phi Scholastic Honor Society. He has received numerous honors and awards, including the 1994 Outstanding Paper Award of the Journal of Information and Education, the 1995 Outstanding Paper Award of the Computer Society of the Republic of China, the 1997 Outstanding Youth Electrical Engineer Award granted by the Chinese Institute of Electrical Engineering (CIEE), Republic of China, the Best Paper Award of the 1999 National Computer Symposium, Republic of China, the 1999 Outstanding Paper Award of the Computer Society of the Republic of China, the 2001 Distinguished Talented Person Award, Republic of China, for Distinguished Contributions in Information Technologies, the 2002 Distinguished Electrical Engineering Professor Award granted by the CIEE, Republic of China, the 2003 Outstanding Paper Award of the Technological and Vocational Education Society, Republic of China, the 2006 Distinguished Research Award of the NTUST, the Outstanding Paper Award of the 2006 11th Conference on Artificial Intelligence and Applications, the Best Paper Award of the 2007 20th International Conference on Industrial, Engineering and Other Applications of Applied Intelligent Systems, the 2008 Distinguished Engineering Professor Award granted by the Chinese Institute of Engineers, Republic of China, the Best Paper Award of the 2009 International Conference on Advanced Information Technologies, the Best Paper Award of the 2009 Fourth International Conference on Innovative Computing, Information and Control, and the Best Paper Award of the 2010 International Conference on Computational Aspects of Social Networks. He is currently the President of the Taiwanese Association for Consumer Electronics, a Co-Chair of the IEEE Systems, Man, and Cybernetics (SMC) Society Technical Committee on Intelligent Internet Systems, the Counselor of the IEEE Student Branch at the NTUST, and the Advisor of the IEEE SMC Student Branch Chapter at the NTUST. He was the President of the Taiwanese Association for Artificial Intelligence from January 2005 to January 2009, an officer of the IEEE Taipei Section from January 2007 to December 2008, the Vice-Chair of the IEEE Computational Intelligence Society, Taipei Chapter, Taiwan, from January 2008 to December 2009, and a member of the Board of Governors of the IEEE Systems, Man, and Cybernetics Society from January 2008 to December 2010. He was a member of the Board of Directors of the Chinese Fuzzy Systems Association from January 1998 to December 2004 and a member of the Board of Governors of the Taiwan Fuzzy Systems Association from January 2008 to December 2010. He is an Associate Editor of the IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS—PART A, an Associate Editor of the IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS—PART C, an Associate Editor of the IEEE TRANSACTIONS ON FUZZY SYSTEMS, an Associate Editor of the IEEE Computational Intelligence Magazine, an Associate Editor of Information Sciences, an Associate Editor of the International Journal of Applied Intelligence, an Associate Editor of the Journal of Intelligent and Fuzzy Systems, an Associate Editor of the International Journal of Artificial Intelligence Tools, an Associate Editor of the International Journal of Pattern Recognition and Artificial Intelligence, an Editorial Board Member of Soft Computing, an Editor of the New Mathematics and Natural Computation Journal, and an Associate Editor of the International Journal of Fuzzy Systems. He is listed in the *International Who's Who of Professionals*, *Marquis Who's Who in the World*, and *Marquis Who's Who in Science and Engineering*.



Huai-Ping Chu received the B.S. degree from the Department of Computer Science, Tung-Hai University, Taichung, Taiwan, in June 1993, and the M.S. degree in computer science and information engineering from National Taiwan University of Science and Technology, Taipei, Taiwan, in June 2010.

He is currently working in the bank's Information Technology department. His current research interests include fuzzy information systems, fuzzy time series forecasting and data mining.



Tian-Wei Sheu received the Ph.D. degree in mathematics from National Osaka University, Osaka, Japan, in 1990.

He is the Dean of the College of Education, National Taichung University of Education, Taichung, Taiwan, and a Professor of the Graduate Institute of Educational Measurement and Statistics, National Taichung University of Education. His studies focused on the item response theory, educational measurement and e-Learning.